

SKY130 Analog IC Physical Verification Quick Sheet

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This document provides a concise and practical reference for full physical verification of analog IC blocks using the SKY130 PDK. The focus is strictly on **DRC**, **LVS**, and **PEX** and how they interact within the post-layout verification flow.

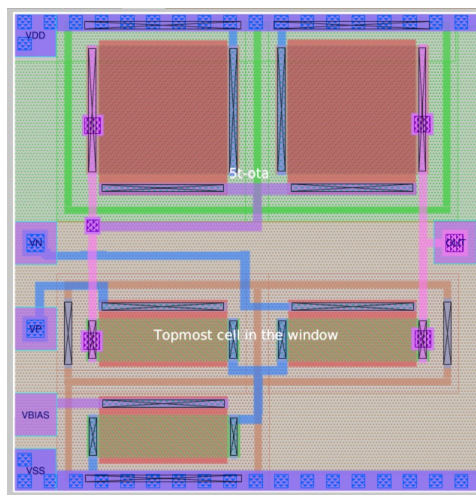


Figure 1: Simple layout of a 5T-OTA

1. Design Rule Check (DRC) – Magic

Goal: ensure the layout complies with foundry spacing/width rules.

Magic tkcon commands:

```
load 5t-ota
drc check
drc count
drc why
drc find
```

When `drc count = 0`, the layout is DRC-clean.

2. Layout Extraction (for LVS and PEX)

Magic:

```
load 5t-ota
extract all
```

3. Layout vs Schematic (LVS) – Netgen

Goal: verify that layout matches schematic connectivity.

```
Contents of circuit 1: Circuit: 'x5t-ota'
Circuit x5t-ota contains 5 device instances.
  Class: sky130_fd_pr_nfet_01v8 instances: 3
  Class: sky130_fd_pr_pfet_01v8 instances: 2
Circuit contains 8 nets.
Contents of circuit 2: Circuit: '5t-ota'
Circuit 5t-ota contains 5 device instances.
  Class: sky130_fd_pr_nfet_01v8 instances: 3
  Class: sky130_fd_pr_pfet_01v8 instances: 2
Circuit contains 8 nets.

Circuit 1 contains 5 devices, Circuit 2 contains 5 devices.
Circuit 1 contains 8 nets, Circuit 2 contains 8 nets.

Final result:
Circuits match uniquely.
Logging to file "lvs_report.out" disabled
LVS Done.
(ota) 51 % |
```

Figure 2: LVS pass message

Magic export:

```
ext2spice lvs
ext2spice
```

Netgen LVS command:

```
lvs "5t-ota.spice 5t-ota" \
    "5t-ota_schematics.spice 5t-ota" \
    /foss/pdks/volare/sky130/.../libs.tech/netgen/sky130A_setup.tcl \
    lvs_report.out
```

Success criterion: *Final result: Circuits match uniquely.*

4. Parasitic Extraction (PEX)

Goal: include post-layout parasitic capacitances and resistances.

Magic tkcon:

```
load 5t-ota
extract all
ext2spice hierarchy on
ext2spice cthresh 0 rthresh 0
ext2spice -o 5t-ota_pex.spice
```

PEX netlist must contain a proper `\subckt` line with the OTA pin order.

5. Post-Layout SPICE Simulation – ngspice

Model libraries must be included **before** the PEX netlist.

```
.lib /foss/pdks/.../sky130.lib.spice tt
.include 5t-ota_pex.spice
```

OTA instantiation (pin order from PEX):

```
XOTA VN VP VSS VBIAS VDD OUT x5t-ota
```

Minimal open-loop AC testbench:

```
VDD VDD 0 1.8
VSS VSS 0 0
VINP VP 0 DC 0.9 AC 1
VINN VN 0 DC 0.9
VBIAS VBIAS 0 0.9
RCL OUT VN 1e9
```

```
XOTA VN VP VSS VBIAS VDD OUT x5t-ota
```

```
.ac dec 100 1 1e9
.print ac mag(v(OUT))
```

Toolchain Logic Summary

Step	Tool	Output	Next
DRC	Magic	Clean layout	Extraction
Extraction	Magic	.ext netlist	LVS / PEX
LVS	Netgen	Match report	PEX
PEX	Magic	_pex.spice	Post-layout sims
Post-layout sim	ngspice	GBW, PM, PSRR, etc.	Sign-off

Final Sign-Off Checklist

- DRC: clean
- LVS: *Circuits match uniquely*
- PEX generated with hierarchy and no thresholds
- Post-layout AC/transient simulations validated
- PSRR checked under PEX
- No long high-impedance poly/LI nets
- Edge dummy fingers for matching

Appendix A — PEX Netlist of the 5T OTA

The following listing shows the full post-layout parasitic extraction (PEX) netlist generated from Magic for the 5-transistor operational transconductance amplifier (OTA) under the SKY130A PDK. This netlist was used for all post-layout simulations (AC, transient, and PSRR).

File: 5t-ota_pex.spice

```
* SPICE3 file created from 5t-ota.ext - technology: sky130A

.subckt sky130_fd_pr__nfet_01v8_CQF6L3 a_n460_n243# a_300_n131# a_n300_n157#
    a_n358_n131#
X0 a_300_n131# a_n300_n157# a_n358_n131# a_n460_n243# sky130_fd_pr__nfet_01v8 ad
    =0.29 pd=2.58 as=0.29 ps=2.58 w=1 l=3
C0 a_n300_n157# a_n358_n131# 0.04241f
C1 a_n358_n131# a_300_n131# 0.02168f
C2 a_n300_n157# a_300_n131# 0.04241f
C3 a_300_n131# a_n460_n243# 0.15607f
C4 a_n358_n131# a_n460_n243# 0.13856f
C5 a_n300_n157# a_n460_n243# 1.37882f
.ends

.subckt sky130_fd_pr__nfet_01v8_LLVQBN a_n460_n243# a_300_n131# a_n300_n157#
    a_n358_n131#
X0 a_300_n131# a_n300_n157# a_n358_n131# a_n460_n243# sky130_fd_pr__nfet_01v8 ad
    =0.29 pd=2.58 as=0.29 ps=2.58 w=1 l=3
C0 a_n358_n131# a_n300_n157# 0.04241f
C1 a_300_n131# a_n300_n157# 0.04241f
C2 a_n358_n131# a_300_n131# 0.02168f
C3 a_300_n131# a_n460_n243# 0.14232f
C4 a_n358_n131# a_n460_n243# 0.14232f
C5 a_n300_n157# a_n460_n243# 1.36096f
.ends

.subckt sky130_fd_pr__pfet_01v8_ZBFWRU a_300_n264# w_n496_n484# a_n358_n264#
    a_n300_n361# VSUBS
X0 a_300_n264# a_n300_n361# a_n358_n264# w_n496_n484# sky130_fd_pr__pfet_01v8 ad
    =0.87 pd=6.58 as=0.87 ps=6.58 w=3 l=3
C0 a_n358_n264# w_n496_n484# 0.05003f
C1 a_300_n264# a_n300_n361# 0.11326f
C2 a_300_n264# w_n496_n484# 0.05003f
C3 a_300_n264# a_n358_n264# 0.06434f
C4 a_n300_n361# w_n496_n484# 0.51232f
C5 a_n358_n264# a_n300_n361# 0.11326f
C6 a_300_n264# VSUBS 0.30902f
C7 a_n358_n264# VSUBS 0.30902f
C8 a_n300_n361# VSUBS 0.84012f
C9 w_n496_n484# VSUBS 3.15269f
.ends

.subckt sky130_fd_pr__nfet_01v8_JRXJ5V a_n460_n243# a_300_n131# a_n300_n157#
    a_n358_n131#
```

```

X0 a_300_n131# a_n300_n157# a_n358_n131# a_n460_n243# sky130_fd_pr__nfet_01v8 ad
    =0.29 pd=2.58 as=0.29 ps=2.58 w=1 l=3
C0 a_300_n131# a_n358_n131# 0.02168f
C1 a_300_n131# a_n300_n157# 0.04241f
C2 a_n358_n131# a_n300_n157# 0.04241f
C3 a_300_n131# a_n460_n243# 0.13856f
C4 a_n358_n131# a_n460_n243# 0.15607f
C5 a_n300_n157# a_n460_n243# 1.37882f
.ends

.subckt x5t-ota VN VP VSS VBIAS VDD OUT
XXM12 VSS OUT VN li_2402_n1715# sky130_fd_pr__nfet_01v8_CQF6L3
XXM13 VSS li_2402_n1715# VBIAS VSS sky130_fd_pr__nfet_01v8_LLVQBN
Xsky130_fd_pr__pfet_01v8_ZBFWRU_0 VDD VDD li_1720_n600# li_1720_n600# VSS
    sky130_fd_pr__pfet_01v8_ZBFWRU
XXM10 OUT VDD VDD li_1720_n600# VSS sky130_fd_pr__pfet_01v8_ZBFWRU
XXM11 VSS li_2402_n1715# VP li_1720_n600# sky130_fd_pr__nfet_01v8_JRXJ5V
C0 VSS VN 0.10291f
C1 OUT VDD 0.22473f
C2 VBIAS VDD 0.00288f
C3 OUT li_2402_n1715# 0.00172f
C4 VBIAS li_2402_n1715# 0.02071f
C5 li_1720_n600# VDD 1.17918f
C6 VP VDD 0.00667f
C7 VDD VN 0.14938f
C8 VSS VDD 0.00523f
C9 li_1720_n600# li_2402_n1715# 0.01045f
C10 VP li_2402_n1715# 0.04489f
C11 li_2402_n1715# VN 0.08824f
C12 VSS li_2402_n1715# 0.08091f
C13 li_1720_n600# OUT 0.22364f
C14 OUT VP 0.00575f
C15 li_1720_n600# VBIAS 0.02341f
C16 OUT VN 0.10317f
C17 VBIAS VP 0.24154f
C18 VSS OUT 0.06037f
C19 VBIAS VN 0
C20 VSS VBIAS 0.48413f
C21 li_1720_n600# VP 0.26294f
C22 li_2402_n1715# VDD 0
C23 li_1720_n600# VN 0.27049f
C24 li_1720_n600# VSS 0.0537f
C25 VP VN 0.22841f
C26 VSS VP 0.0889f
C27 OUT 0 0.90444f
C28 VDD 0 6.51335f
C29 li_1720_n600# 0 2.00947f
C30 VP 0 1.2526f
C31 li_2402_n1715# 0 0.32877f
C32 VSS 0 1.34994f
C33 VBIAS 0 1.19081f
C34 VN 0 1.57073f
.ends

```